



Computer Networks (NET 225) Assignment Duration: 2 weeks

Due Date: 13th June 2026

Programme: BMIS

- a. Using a real-life example in Malawi (i.e. provide an example for each), discuss in detail what the following are and how they are used:
- i. Network services and Broadband technologies
 - ii. Value Added Networks
 - iii. Packet switching and Circuit switching
 - iv. Frame relay, asynchronous transfer mode
 - v. Integrated service digital network DSL
 - vi. Cable modem and T1 line
 - vii. Convergence network and unified messaging
 - viii. The electronic spectrum
 - ix. Private branch exchange
 - x. Narrowband, broadband and access providers

50 marks

- b. With the aid of a well-labelled, simple TCP/IP layered model(s) diagram briefly explain how a data packet is encapsulated from a source host and de-capsulated at a destination host

20 marks

- c. Network topology refers to the arrangement of elements within a network. Like network diagrams, network topologies can describe either the physical or logical aspects of a network. Cisco Packet Tracer is a powerful network simulation software from Cisco Systems Inc which can simulate/create a network without having a physical network. It has a drag and drop interface that is easy to use yet highly effective while configuring complex networks.

Required:

- i. Using Cisco Packet Tracer, design a simple logical network topology for a computer Lab with 20 workstations (include all required network hardware in your design)

15 marks

- ii. Using Cisco Packet Tracer, design a physical physical network topology for a computer Lab with 20 workstations (include all required network hardware in your design)

15 marks

Total: 100 marks

(Plagiarism is heavily penalized)

How to Submit

- Hand in your assignment file to the assessor as follows:
 - ⇒ Indicate your Name, Reg# and other programme identification details on the cover page of your assignment
 - ⇒ Use Harvard referencing style
 - ⇒ Send the **Zipped folder** to kchimphanda@mca.ac.mw by **13th June 2026 11:59 PM**
 - ⇒ **Late submissions will not be marked**

End of Assignment

***** Good Luck *****